

Statistics (2)

Syllabus

Yu-You Liou

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Lecturer: Yu-You Liou	Contact: d10627008@ntu.edu.tw
Course No.: EIB-21F-01-A1	Website: https://yyliou.github.io/stat
Meeting Time: Thursday, Periods 6–7 (13:10–15:00)	Location: L309

Course Objectives

1. To train students in basic statistical knowledge and econometric concepts.
2. To enhance students' familiarity with data analysis and provide hands-on experience in statistical analysis.

Textbooks

1. Bluman, A. G. (2012). *Elementary Statistics: A Step-by-Step Approach* (8th ed.). McGraw-Hill.

Course Structure

1. **Lecture** (First hour): Foundational knowledge and concepts introduced each session.
2. **Lab** (Second hour): Hands-on in-class exercises using R to apply the concepts learned during the lecture.

Assessment Criteria

1. **In-Class Exercises** (36%): 12 exercises $\times$ 3% each, submitted at the end of each session.
2. **Mid-Term Exam** (32%) and **Final Exam** (32%).

Regulations

1. Attendance

Per Shih Chien University regulations, students who are absent for more than one-third of class sessions will receive a final grade of zero. Make-up attendance will not be permitted.

2. Examination

Students who do not show up for their scheduled examination will receive a grade of zero for that component.

3. In-class exercise deadline

The deadline for all in-class exercises is before midnight of the following week. Late submissions will not be accepted.

4. Academic Integrity

All submitted work must be the student's own. Copying from other students is strictly prohibited. AI detection tools are used to review submissions; any work found to be copied will receive a grade of zero.

5. AI Policy

Students may use AI tools (e.g., ChatGPT, GitHub Copilot) as learning aids, but must be able to explain and defend any code or analysis they submit. Uncritical submission of AI-generated work without understanding constitutes academic dishonesty.

Required Software

All software is free and open-source: **R** at <https://cran.r-project.org>, **RStudio Desktop** at <https://posit.co/download/rstudio-desktop>, and R packages **tidyverse**, **ggplot2**, **patchwork** (installation instructions on the course R page).

Course Schedule

Week	Topic
1	Explain the syllabus and class regulations
2	Chapter 8: Hypothesis testing (ex1)
3	Chapter 8: Hypothesis testing (ex2)
4	Chapter 8: Hypothesis testing (ex3)
5	Chapter 9: Testing the difference between two means, two proportions, and two variances (ex4)
6	Chapter 9: Testing the difference between two means, two proportions, and two variances (ex5)
7	Chapter 10: Correlation and regressions (ex6)
8	Midterm review
9	Midterm exam
10	Chapter 10: Correlation and regression (ex7)
11	Chapter 10: Correlation and regression (ex8)
12	Chapter 11: Other chi-square tests (ex9)
13	Chapter 11: Other chi-square tests (ex10)
14	Chapter 12: Analysis of variance (ex11)
15	Chapter 12: Analysis of variance (ex12)
16	Final exam
17	Flexible learning week
18	Flexible learning week

Note: Syllabus is subject to change. Any updates will be announced on the course website.